

5 $\lim_{z \rightarrow 1^+} f(z) = N$

6 $\lim_{x \rightarrow c^+} s(x) = D$

Exer. 7–12: For the given $\lim_{x \rightarrow a} f(x) = L$ and ϵ , use the graph of f to find the largest δ such that if $0 < |x - a| < \delta$, then $|f(x) - L| < \epsilon$.

7 $\lim_{x \rightarrow 3/2} \frac{4x^2 - 9}{2x - 3} = 6; \quad \epsilon = 0.01$

8 $\lim_{x \rightarrow -2/3} \frac{9x^2 - 4}{3x + 2} = -4; \quad \epsilon = 0.1$

9 $\lim_{x \rightarrow 4} x^2 = 16; \quad \epsilon = 0.1$

10 $\lim_{x \rightarrow 3} x^3 = 27; \quad \epsilon = 0.01$

11 $\lim_{x \rightarrow 16} \sqrt{x} = 4; \quad \epsilon = 0.1$

12 $\lim_{x \rightarrow 27} \sqrt[3]{x} = 3; \quad \epsilon = 0.1$

Exer. 13–24: Use Definition (2.4) to prove that the limit exists.

13 $\lim_{x \rightarrow 3} 5x = 15$

14 $\lim_{x \rightarrow 5} (-4x) = -20$

15 $\lim_{x \rightarrow -3} (2x + 1) = -5$

16 $\lim_{x \rightarrow 2} (5x - 3) = 7$

17 $\lim_{x \rightarrow -6} (10 - 9x) = 64$

18 $\lim_{x \rightarrow 4} (15 - 8x) = -17$

19 $\lim_{x \rightarrow 6} (3 - \frac{1}{2}x) = 0$

20 $\lim_{x \rightarrow 6} (9 - \frac{1}{6}x) = 8$

21 $\lim_{x \rightarrow 3} 5 = 5$

22 $\lim_{x \rightarrow 5} 3 = 3$

23 $\lim_{x \rightarrow a} c = c$ for all real numbers a and c

24 $\lim_{x \rightarrow a} (mx + b) = ma + b$ for all real numbers m , b , and a

Exer. 25–30: Use the graphical method illustrated in Example 2 to verify the limit for $a > 0$.

25 $\lim_{x \rightarrow -a} x^2 = a^2$

26 $\lim_{x \rightarrow a} (x^2 + 1) = a^2 + 1$

27 $\lim_{x \rightarrow a} x^3 = a^3$

28 $\lim_{x \rightarrow a} x^4 = a^4$

29 $\lim_{x \rightarrow a} \sqrt{x} = \sqrt{a}$

30 $\lim_{x \rightarrow a} \sqrt[3]{x} = \sqrt[3]{a}$

Exer. 31–38: Use the method illustrated in Examples 3 and 4 to show that the limit does not exist.

31 $\lim_{x \rightarrow 3} \frac{|x - 3|}{x - 3}$

32 $\lim_{x \rightarrow -2} \frac{x + 2}{|x + 2|}$

33 $\lim_{x \rightarrow -1} \frac{3x + 3}{|x + 1|}$

34 $\lim_{x \rightarrow 5} \frac{2x - 10}{|x - 5|}$

35 $\lim_{x \rightarrow 0} \frac{1}{x^2}$

36 $\lim_{x \rightarrow 4} \frac{7}{x - 4}$

37 $\lim_{x \rightarrow -5} \frac{1}{x + 5}$

38 $\lim_{x \rightarrow 1} \frac{1}{(x - 1)^2}$

39 Give an example of a function f that is defined at a such that $\lim_{x \rightarrow a} f(x)$ exists and $\lim_{x \rightarrow a} f(x) \neq f(a)$.

40 If f is the greatest integer function (see Example 4 of Section 1.2) and a is any integer, show that $\lim_{x \rightarrow a} f(x)$ does not exist.

41 Let f be defined as follows:

$$f(x) = \begin{cases} 0 & \text{if } x \text{ is rational} \\ 1 & \text{if } x \text{ is irrational} \end{cases}$$

Prove that for every real number a , $\lim_{x \rightarrow a} f(x)$ does not exist.

42 Why is it impossible to investigate $\lim_{x \rightarrow 0} \sqrt{x}$ by means of Definition (2.4)?

2.3 TECHNIQUES FOR FINDING LIMITS

It would be an excruciating task to verify every limit by means of Definition (2.4) or (2.5). The purpose of this section is to introduce theorems that may be used to simplify problems involving limits. Before stating the first theorem, let us consider the limits of two very simple functions:

(i) The constant function f given by $f(x) = c$

(ii) The linear function g given by $g(x) = x$

The graph of f is the horizontal line $y = c$ sketched in Figure 2.19 for the case $c > 0$. Since

$$|f(x) - c| = |c - c| = 0 \quad \text{for every } x$$

imation of L , and let
value of W . These errors

dL .

metric tons.

$$\frac{1.18}{1.8} \frac{dL}{L} = \frac{3.18}{L} \frac{dL}{L}$$

20.

$\pm 20\%$.

importance to pilots
and speed v at a height
a differentiable func-
 dh (the instantaneous
possible to know the
be estimated by using
the situation illustrated
 v_0 and v_1 at heights
at height h_1 may be

ained by observation
value $P = \frac{1}{4}$ is some

bove the ground the
ussion (with $P = \frac{1}{4}$)
round.

discussion, we let

0.

EXERCISES 3.5

135

Solving $(v_0/v_1) = (h_0/h_1)^P$ for v_1 and then substituting values, we obtain

$$v_1 = v_0 \left(\frac{h_1}{h_0} \right)^P = 28 \left(\frac{200}{20} \right)^{1/7} \approx 39 \text{ mi/hr.}$$

At $h_1 = 200$,

$$\left. \frac{dv}{dh} \right|_{h=h_1} \approx \frac{v_1 - v_0}{h_1 - h_0} \approx \frac{39 - 28}{200 - 20} \approx 0.06.$$

Thus, at a height of 200 feet, the vertical wind shear is approximately 0.06 (mi/hr)/ft. This is a common value. Wind shear values greater than 0.1 are considered high.

EXERCISES 3.5

Exer. 1–4: (a) Find general formulas for Δy and dy .
(b) If, for the given values of a and Δx , x changes from a to $a + \Delta x$, find the values of Δy and dy .

1 $y = 2x^2 - 4x + 5$; $a = 2$, $\Delta x = -0.2$

2 $y = x^3 - 4$; $a = -1$, $\Delta x = 0.1$

3 $y = 1/x^2$; $a = 3$, $\Delta x = 0.3$

4 $y = \frac{1}{2+x}$; $a = 0$, $\Delta x = -0.03$

Exer. 5–10: Find (a) Δy , (b) dy , and (c) $dy - \Delta y$.

5 $y = 4 - 9x$

6 $y = 7x + 12$

7 $y = 3x^2 + 5x - 2$

8 $y = 4 - 7x - 2x^2$

9 $y = 1/x$

10 $y = 1/x^2$

Exer. 11–18: Find a linear approximation for $f(b)$ if the independent variable changes from a to b .

11 $f(x) = 4x^5 - 6x^4 + 3x^2 - 5$; $a = 1$, $b = 1.03$

12 $f(x) = -3x^3 + 8x - 7$; $a = 4$, $b = 3.96$

13 $f(x) = x^4$; $a = 1$, $b = 0.98$

14 $f(x) = x^4 - 3x^3 + 4x^2 - 5$; $a = 2$, $b = 2.01$

15 $f(\theta) = 2 \sin \theta + \cos \theta$; $a = 30^\circ$, $b = 27^\circ$

16 $f(\phi) = \csc \phi + \cot \phi$; $a = 45^\circ$, $b = 46^\circ$

17 $f(z) = \sec z$; $a = 60^\circ$, $b = 62^\circ$

18 $f(\beta) = \tan \beta$; $a = 30^\circ$, $b = 28^\circ$

19 (a) If $f(x) = \sin(\tan x - 1)$, find an (approximate) equation of the tangent line to the graph of f at $(2.5, f(2.5))$, using Exercise 51 in Section 3.2.

(b) Use the equation found in (a) to approximate $f(2.6)$.

(c) Use (3.31) with $x = 2.5$ to approximate $f(2.6)$.

(d) Compare the two approximations in (b) and (c).

20 (a) If $f(x) = x^3 + 3x^2 - 2x + 5$, find an (approximate) equation of the tangent line to the graph of f at $(0.4, f(0.4))$.

(b) Use the equation found in (a) to approximate $f(0.43)$.

(c) Use (3.31) with $x = 0.4$ to approximate $f(0.43)$.

(d) Compare the two approximations in (b) and (c).

Exer. 21–24: Let x denote a measurement with a maximum error of Δx . Use differentials to approximate the average error and the percentage error for the calculated value of y .

21 $y = 3x^4$; $x = 2$, $\Delta x = \pm 0.01$

22 $y = x^3 + 5x$; $x = 1$, $\Delta x = \pm 0.1$

23 $y = 4\sqrt{x} + 3x$; $x = 4$, $\Delta x = \pm 0.2$

24 $y = 6\sqrt[3]{x}$; $x = 8$, $\Delta x = \pm 0.03$

25 If $A = 3x^2 - x$, find dA for $x = 2$ and $dx = 0.1$.

26 If $P = 6t^{2/3} + t^2$, find dP for $t = 8$ and $dt = 0.2$.

27 If $y = 4x^3$ and the maximum percentage error in x is $\pm 15\%$, approximate the maximum percentage error in y .

28 If $z = 40\sqrt{w^2}$ and the maximum average error in w is ± 0.08 , approximate the maximum average error in z .

29 If $A = 15\sqrt[3]{s^2}$ and the allowable maximum average error in A is to be ± 0.04 , determine the allowable maximum average error in s .

30 If $S = 10\pi x^2$ and the allowable maximum percentage error in S is to be $\pm 10\%$, determine the allowable maximum percentage error in x .

31 The radius of a circular manhole cover is estimated to be 16 inches, with a maximum error in measurement of ± 0.06 inch. Use differentials to estimate the maximum error in the calculated area of one side of the cover. Approximate the average error and the percentage error.

- 32 The length of a side of a square floor tile is estimated as 1 foot, with a maximum error in measurement of $\pm \frac{1}{16}$ inch. Use differentials to estimate the maximum error in the calculated area. Approximate the average error and the percentage error.
- 33 Use differentials to approximate the increase in volume of a cube if the length of each edge changes from 10 inches to 10.1 inches. What is the exact change in volume?
- 34 A spherical balloon is being inflated with gas. Use differentials to approximate the increase in surface area of the balloon if the diameter changes from 2 feet to 2.02 feet.
- 35 One side of a house has the shape of a square surmounted by an equilateral triangle. If the length of the base is measured as 48 feet, with a maximum error in measurement of ± 1 inch, calculate the area of the side. Use differentials to estimate the maximum error in the calculation. Approximate the average error and the percentage error.
- 36 Small errors in measurements of dimensions of large containers can have a marked effect on calculated volumes. A silo has the shape of a right circular cylinder surmounted by a hemisphere (see figure). The altitude of the cylinder is exactly 50 feet. The circumference of the base is measured as 30 feet, with a maximum error in measurement of ± 6 inches. Calculate the volume of the silo from these measurements, and use differentials to estimate the maximum error in the calculation. Approximate the average error and the percentage error.

EXERCISE 36



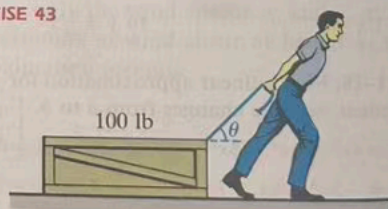
- 37 As sand leaks out of a container, it forms a conical pile whose altitude is always the same as the radius. If, at a certain instant, the radius is 10 centimeters, use differentials to approximate the change in radius that will increase the volume of the pile by 2 cm^3 .
- 38 An isosceles triangle has equal sides of length 12 inches. If the angle θ between these sides is increased from 30° to 33° , use differentials to approximate the change in the area of the triangle.
- 39 Newton's law of gravitation states that the force F of attraction between two particles having masses m_1 and m_2 is given by $F = Gm_1m_2/s^2$, where G is a constant and s is the distance between the particles. If $s = 20 \text{ cm}$, use differentials to approximate the change in s that will increase F by 10% .

- 40 The formula $T = 2\pi\sqrt{l/g}$ relates the length l of a pendulum to its period T , where g is a gravitational constant. What percentage change in the length corresponds to a 30% increase in the period?
- 41 Constriction of arterioles is a cause of high blood pressure. It has been verified experimentally that as blood flows through an arteriole of fixed length, the pressure difference between the two ends of the arteriole is inversely proportional to the fourth power of the radius. If the radius of an arteriole decreases by 10% , use differentials to find the percentage change in the pressure difference.
- 42 The electrical resistance R of a wire is directly proportional to its length and inversely proportional to the square of its diameter. If the length is fixed, how accurately must the diameter be measured (in terms of percentage error) to keep the percentage error in R between -3% and 3% ?
- 43 If an object of weight W pounds is pulled along a horizontal plane by a force applied to a rope that is attached to the object and if the rope makes an angle θ with the horizontal, then the magnitude of the force is given by

$$F(\theta) = \frac{\mu W}{\mu \sin \theta + \cos \theta},$$

where μ is a constant called the *coefficient of friction*. Suppose that a 100-pound box is being pulled along a floor and that $\mu = 0.2$ (see figure). If θ is changed from 45° to 46° , use differentials to approximate the change in the force that must be applied.

EXERCISE 43



- 44 It will be shown in Chapter 15 that if a projectile is fired from a cannon with an initial velocity v_0 and at an angle α to the horizontal, then its maximum height h and range R are given by

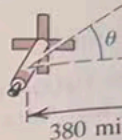
$$h = \frac{v_0^2 \sin^2 \alpha}{2g} \quad \text{and} \quad R = \frac{2v_0^2 \sin \alpha \cos \alpha}{g}.$$

Suppose $v_0 = 100 \text{ ft/sec}$ and $g = 32 \text{ ft/sec}^2$. If α is increased from 30° to $30^\circ 30'$, use differentials to estimate the changes in h and R .

- 45 At a point 20 feet from the base of a flagpole, the angle of elevation of the top of the pole is measured as 60° , with a possible error of $\pm 15'$. Use differentials to approximate the error in the calculated height of the pole.

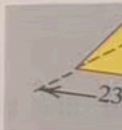
- 46 A spacelab circles Earth. When an astronaut in the figure is $65.8 \pm 0.5^\circ$. Use differentials to approximate the astronaut's calculation.

EXERCISE 46



- 47 The Great Pyramid of Giza is one of the most massive structures ever built. One of the sides of the pyramid is measured to be 230 ± 1 meters. Use differentials to approximate the error in the calculated height of the pyramid.

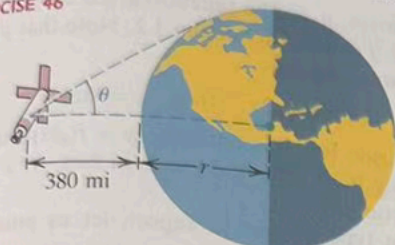
EXERCISE 47



- 48 As a point light source is shown in the figure, it is inversely proportional to the distance from the light source to the object.

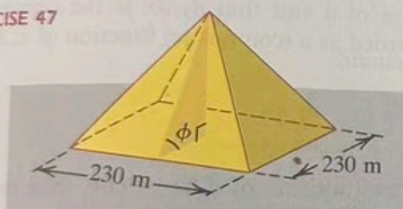
- 45 A spacelab circles the earth at an altitude of 380 miles. When an astronaut views the horizon, the angle θ shown in the figure is 65.8° , with a possible maximum error of $\pm 0.5^\circ$. Use differentials to approximate the error in the astronaut's calculation of the radius of the earth.

EXERCISE 46



- 47 The Great Pyramid of Egypt has a square base of 230 meters (see figure). To estimate the height h of this massive structure, an observer stands at the midpoint of one of the sides and views the apex of the pyramid. The angle of elevation ϕ is found to be 52° . How accurate must this measurement be to keep the error in h between -1 meter and 1 meter?

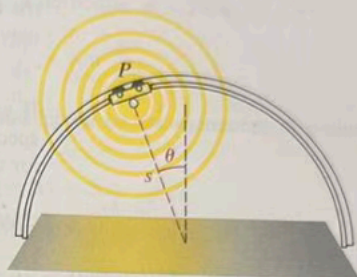
EXERCISE 47



- 48 As a point light source moves on a semicircular track, as shown in the figure, the illuminance E on the surface is inversely proportional to the square of the distance s

from the source and is directly proportional to the cosine of the angle θ between the direction of light flow and the normal to the surface. If θ is decreased from 21° to 20° and s is constant, use differentials to approximate the percentage increase in illuminance.

EXERCISE 48



- 49 Boyle's law states that if the temperature is constant, the pressure p and volume v of a confined gas are related by the formula $pv = c$, where c is a constant or, equivalently, by $p = c/v$ with $v \neq 0$. Show that dp and dv are related by means of the formula $p dv + v dp = 0$.
- 50 In electrical theory Ohm's law states that $I = V/R$, where I is the current (in amperes), V is the electromotive force (in volts), and R is the resistance (in ohms). Show that dI and dR are related by the formula $R dI + I dR = 0$.
- 51 The area A of a square of side s is given by $A = s^2$. If s increases by an amount Δs , illustrate dA and $\Delta A - dA$ geometrically.
- 52 The volume V of a cube of edge s is given by $V = s^3$. If s increases by an amount Δs , illustrate dV and $\Delta V - dV$ geometrically.

3.6 THE CHAIN RULE

The rules for derivatives obtained in previous sections are limited in scope because they can be used only for sums, differences, products, and quotients that involve x^n , $\sin x$, $\cos x$, $\tan x$, and so on. There is no rule that can be applied directly to expressions such as $\sin 2x$ or $\sqrt{x^2 + 1}$. Note that

$$D_x \sin 2x \neq \cos 2x,$$

for if we use the identity $\sin 2x = 2 \sin x \cos x$ and apply the product rule,

$$\begin{aligned} D_x \sin 2x &= D_x (2 \sin x \cos x) \\ &= 2 D_x (\sin x \cos x) \\ &= 2 [\sin x (D_x \cos x) + \cos x (D_x \sin x)] \\ &= 2 [\sin x (-\sin x) + \cos x (\cos x)] \\ &= 2 (-\sin^2 x + \cos^2 x) \\ &= 2 \cos 2x. \end{aligned}$$

...ATIONS OF THE DERIVATIVE
... follows:

... $4)^{1/3}$

... $f''(x)$ does not exist
... $\frac{1}{7}$, and 4.

... critical numbers of f

... $\sin x = \frac{1}{2}$ or
... $[0, 2\pi]$ are $\pi/6$,
... sketch the graph

... emum at c . If
... exists, then

... occur. Thus,

... containing

... es that if
... positive

EXERCISES 4.1

175

or both negative;

$$f(x) - f(c) < 0 \quad \text{whenever } x - c < 0$$

$$f(x) - f(c) > 0 \quad \text{whenever } x - c > 0$$

Equivalently, if x is in (a, b) and $x \neq c$:

$$f(x) < f(c) \quad \text{whenever } x < c$$

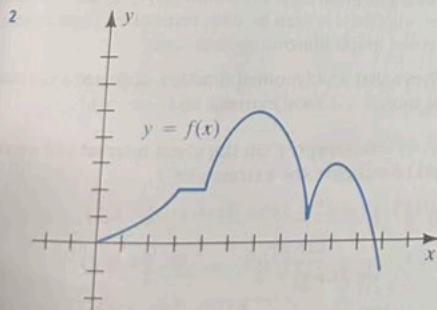
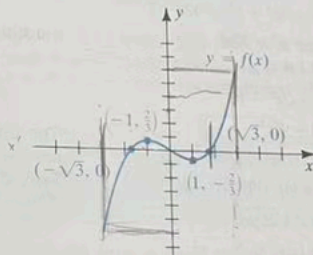
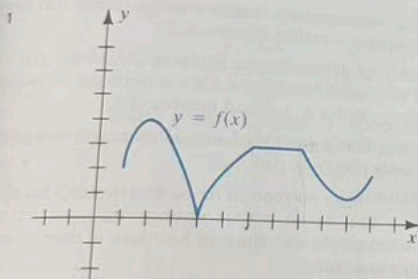
$$f(x) > f(c) \quad \text{whenever } x > c$$

It follows that $f(c)$ is neither a local maximum nor a local minimum for f , contrary to hypothesis. Consequently, (i) cannot occur. Similarly, the assumption that $f'(c) < 0$ leads to a contradiction. Hence (iii) must hold and the theorem is proved. ■

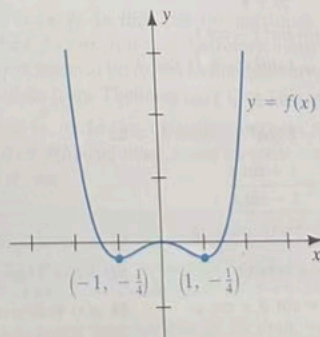
EXERCISES 4.1

Exer. 1-2: Use the graph to estimate the absolute maximum, the absolute minimum, and the local extrema of f .

- 3 (a) $[-7, 3]$ (b) $(-3, \sqrt{3})$ (c) $[-\sqrt{3}, 1]$ (d) $[0, 3]$



- 4 (a) $[-2, 2]$ (b) $(0, 2)$ (c) $(-1, 1)$ (d) $[-2, -1]$



Exer. 3-4: Use the graph to estimate the extrema of f on each interval.

Exer. 5–6: Sketch the graph of f , and find the extrema on each interval.

5 $f(x) = \frac{1}{3}x^3 - 2x$

(a) $[0, 5]$ (b) $(0, 2)$ (c) $(0, 4)$ (d) $[2, 5]$

6 $f(x) = (x-1)^{2/3} - 4$

(a) $[0, 9]$ (b) $(1, 2)$ (c) $(-7, 2)$ (d) $[0, 1]$

Exer. 7–10: Find the extrema of f on the given interval.

7 $f(x) = 5 - 6x^2 - 2x^3$; $[-3, 1]$

8 $f(x) = 3x^2 - 10x + 7$; $[-1, 3]$

9 $f(x) = 1 - x^{2/3}$; $[-1, 8]$

10 $f(x) = x^4 - 5x^2 + 4$; $[0, 2]$

Exer. 11–36: Find the critical numbers of the function.

11 $f(x) = 4x^2 - 3x + 2$

12 $g(x) = 2x + 5$

13 $s(t) = 2t^3 + t^2 - 20t + 4$

14 $K(z) = 4z^3 + 5z^2 - 42z + 7$

15 $F(w) = w^4 - 32w$

16 $k(r) = r^5 - 2r^3 + r - 12$

17 $f(z) = \sqrt{z^2 - 16}$

18 $M(x) = \sqrt[3]{x^2 - x - 2}$

19 $h(x) = (2x - 5)\sqrt{x^2 - 4}$

20 $T(v) = (4v + 1)\sqrt{v^2 - 16}$

21 $g(t) = t^2 \sqrt[3]{2t - 5}$

22 $g(x) = (x - 3)\sqrt{9 - x^2}$

23 $G(x) = \frac{2x - 3}{x^2 - 9}$

24 $f(s) = \frac{s^2}{5s + 4}$

25 $f(t) = \sin^2 t - \cos t$

26 $g(t) = 4 \sin^3 t + 3\sqrt{2} \cos^2 t$

27 $K(\theta) = \sin 2\theta + 2 \cos \theta$

28 $f(x) = 8 \cos^3 x - 3 \sin 2x - 6x$

29 $f(x) = \frac{1 + \sin x}{1 - \sin x}$

30 $g(\theta) = 2\sqrt{3}\theta + \sin 4\theta$

31 $k(u) = u - \tan u$

33 $H(\phi) = \cot \phi + \csc \phi$

35 $f(x) = \sec(x^2 + 1)$

32 $p(z) = 3 \tan z - 4z$

34 $g(x) = 2x + \cot x$

36 $s(t) = \frac{\sec t + 1}{\sec t - 1}$

37 (a) If $f(x) = x^{1/3}$, prove that 0 is the only critical number of f and that $f(0)$ is not a local extremum.

(b) If $f(x) = x^{2/3}$, prove that 0 is the only critical number of f and that $f(0)$ is a local minimum of f .

38 If $f(x) = |x|$, prove that 0 is the only critical number of f , that $f(0)$ is a local minimum of f , and that the graph of f has no tangent line at the point $(0, 0)$.

Exer. 39–40: (a) Prove that f has no local extrema. (b) Sketch the graph of f . (c) Prove that f is continuous on the interval $(0, 1)$, but f has neither a maximum nor a minimum value on $(0, 1)$. (d) Explain why (c) does not contradict Theorem (4.3).

39 $f(x) = x^3 + 1$

40 $f(x) = \frac{1}{x^2}$

41 (a) Prove that a polynomial function f of degree 1 has no extrema on the interval $(-\infty, \infty)$.

(b) Discuss the extrema of f on a closed interval $[a, b]$.

42 If f is a constant function and (a, b) is any open interval, prove that $f(c)$ is both a local and an absolute extremum of f for every number c in (a, b) .

43 If f is the greatest integer function, prove that every number is a critical number of f .

44 Let f be defined by the following conditions: $f(x) = 0$ if x is rational and $f(x) = 1$ if x is irrational. Prove that every number is a critical number of f .

45 Prove that a quadratic function has exactly one critical number on $(-\infty, \infty)$.

46 Prove that a polynomial function of degree 3 has either two, one, or no critical numbers on $(-\infty, \infty)$, and sketch graphs that illustrate how each of these possibilities can occur.

47 Let $f(x) = x^n$ for a positive integer n . Prove that f has either one or no local extremum on $(-\infty, \infty)$ depending on whether n is even or odd, respectively, and sketch a typical graph illustrating each case.

48 Prove that a polynomial function of degree n can have at most $n - 1$ local extrema on $(-\infty, \infty)$.

[c] Exer. 49–50: Graph f on the given interval and use the graph to estimate the extrema of f .

49 $f(x) = x^6 - x^5 + 3x^3 - 2x + 1$; $[-1, 1]$

50 $f(x) = \frac{x^2 - \cos 3x}{\sin 2x - x + 2}$; $[-\pi, 1]$

[c] Exer. 51–52: Graph f on the given interval and use the graph to estimate the critical numbers of f .

51 $f(x) = |\frac{1}{3}x^3 + x^2 - x - 1.5|$; $[-3, 2]$

52 $f(x) = x \sin^2 x \cos^2 x$; $[-2, 2]$



velocity of P during the time interval $[a, b]$ is

$$v_{av} = \frac{s(b) - s(a)}{b - a}.$$

According to the mean value theorem, this average (mean) velocity is equal to the velocity $s'(c)$ at some time c between a and b .

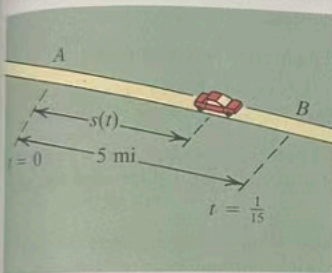
EXAMPLE 4 The speedometer of an automobile registers 50 mi/hr as it passes a mileage marker along a highway. Four minutes later, as the automobile passes a marker that is five miles from the first, the speedometer registers 55 mi/hr. Use the mean value theorem to prove that the velocity exceeded 70 mi/hr at some time while the automobile was traveling between the two markers.

SOLUTION We may assume that the automobile is moving along a straight highway with the two mileage markers at A and B , as illustrated in Figure 4.17. Let t denote the elapsed time (in hours) after the automobile passes A , and let $s(t)$ denote its distance from A (in miles) at time t . Since the automobile passes B at $t = \frac{4}{60} = \frac{1}{15}$ hr, the average velocity during the trip from A to B is

$$v_{av} = \frac{s(\frac{1}{15}) - s(0)}{\frac{1}{15} - 0} = \frac{5}{\frac{1}{15}} = 75 \text{ mi/hr}.$$

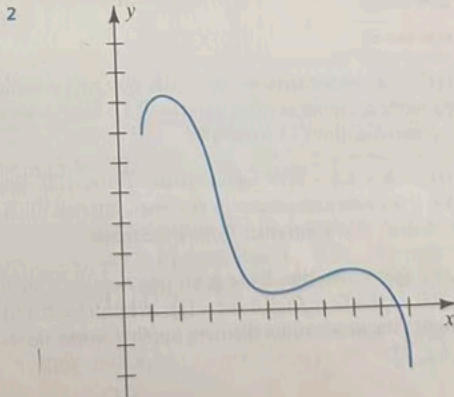
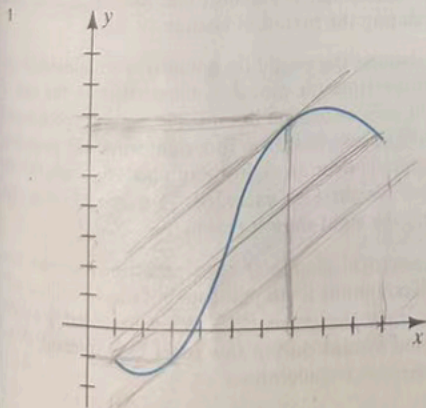
If we assume that s is a differentiable function, then applying the mean value theorem (4.12) to s , with $a = 0$ and $b = \frac{1}{15}$, proves that there is a time c in the time interval $[0, \frac{1}{15}]$ at which the velocity $s'(c)$ of the automobile is 75 mi/hr. Note that the velocity at A or B is irrelevant.

FIGURE 4.17



EXERCISES 4.2

Exer. 1–2: Use the graph of f to estimate the numbers in $[1, 10]$ that satisfy the conclusion of the mean value theorem.



Exer. 3–8: Show that f satisfies the hypotheses of Rolle's theorem on $[a, b]$, and find all numbers c in (a, b) such that $f'(c) = 0$.

3 $f(x) = 3x^2 - 12x + 11$; $[0, 4]$

4 $f(x) = 5 - 12x - 2x^2$; $[-7, 1]$

5 $f(x) = x^4 + 4x^2 + 1$; $[-3, 3] \rightarrow$

6 $f(x) = x^3 - x$; $[-1, 1]$

7 $f(x) = \sin 2x$; $[0, \pi]$

8 $f(x) = \cos 2x + 2 \cos x$; $[0, 2\pi]$

Exer. 9–24: Determine whether f satisfies the hypotheses of the mean value theorem on $[a, b]$, and, if so, find all numbers c in (a, b) such that

$$f(b) - f(a) = f'(c)(b - a).$$

9 $f(x) = 5x^2 - 3x + 1$; $[1, 3] \rightarrow$

10 $f(x) = 3x^2 + x - 4$; $[1, 5]$

11 $f(x) = \frac{1}{(x-1)^2}$; $[0, 2]$

12 $f(x) = \frac{x+3}{x-2}$; $[-2, 3]$

13 $f(x) = x^{2/3}$; $[-8, 8]$

14 $f(x) = |x - 3|$; $[-1, 4]$

15 $f(x) = x + (4/x)$; $[1, 4]$

16 $f(x) = 3x^5 + 5x^3 + 15x$; $[-1, 1]$

17 $f(x) = x^3 - 2x^2 + x + 3$; $[-1, 1]$

18 $f(x) = 1 - 3x^{1/3}$; $[-8, -1]$

19 $f(x) = 4 + \sqrt{x-1}$; $[1, 5]$

20 $f(x) = (x+2)^{2/3}$; $[-1, 6]$

21 $f(x) = x^3 + 1$; $[-2, 4]$

22 $f(x) = x^3 + 4x$; $[-3, 6]$

23 $f(x) = \sin x$; $[0, \pi/2]$

24 $f(x) = \tan x$; $[0, \pi/4]$

25 If $f(x) = |x|$, show that $f(-1) = f(1)$ but $f'(c) \neq 0$ for every number c in the open interval $(-1, 1)$. Why doesn't this contradict Rolle's theorem?

26 If $f(x) = 5 + 3(x-1)^{2/3}$, show that $f(0) = f(2)$ but $f'(c) \neq 0$ for every number c in the open interval $(0, 2)$. Why doesn't this contradict Rolle's theorem?

27 If $f(x) = 4/x$, prove that there is no real number c such that $f(4) - f(-1) = f'(c)[4 - (-1)]$. Why doesn't this contradict the mean value theorem applied to the interval $[-1, 4]$?

28 If f is the greatest integer function and if a and b are real numbers such that $b - a \geq 1$, prove that there is no real number c such that $f(b) - f(a) = f'(c)(b - a)$. Why doesn't this contradict the mean value theorem?

29 If f is a linear function, prove that f satisfies the hypotheses of the mean value theorem on every closed interval $[a, b]$ and that every number c satisfies the conclusion of the theorem.

30 If f is a quadratic function and $[a, b]$ is any closed interval, prove that precisely one number c in the interval (a, b) satisfies the conclusion of the mean value theorem.

31 If f is a polynomial function of degree 3 and $[a, b]$ is any closed interval, prove that at most two numbers in (a, b) satisfy the conclusion of the mean value theorem. Generalize to polynomial functions of degree n for any positive integer n .

32 If $f(x)$ is a polynomial of degree 3, use Rolle's theorem to prove that f has at most three real zeros. Extend this result to polynomials of degree n .

Exer. 33–40: Use the mean value theorem.

33 If f is continuous on $[a, b]$ and if $f'(x) = 0$ for every x in (a, b) , prove that $f(x) = k$ for some real number k .

34 If f is continuous on $[a, b]$ and if $f'(x) = c$ for every x in (a, b) , prove that $f(x) = cx + d$ for some real number d .

35 A straight highway 50 miles long connects two cities A and B. Prove that it is impossible to travel from A to B by automobile in exactly one hour without having the speedometer register 50 mi/hr at least once.

36 Let T denote the temperature (in $^{\circ}\text{F}$) at time t (in hours). If the temperature is decreasing, then dT/dt is the rate of cooling. The greatest temperature variation during a twelve-hour period occurred in Montana in 1916, when the temperature dropped from 44°F to a chilling -56°F . Show that the rate of cooling exceeded -8°F/hr at some time during the period of change.

37 If W denotes the weight (in pounds) of an individual and t denotes time (in months), then dW/dt is the rate of weight gain or loss (in lb/mo). The current speed record for weight loss is a drop in weight from 487 pounds to 130 pounds over an eight-month period. Show that the rate of weight loss exceeded 44 lb/mo at some time during the eight-month period.

38 The electrical charge Q on a capacitor increases from 2 millicoulombs to 10 millicoulombs in 15 milliseconds. Show that the current $I = dQ/dt$ exceeded $\frac{1}{2}$ ampere at some instant during this short time interval. (Note: 1 ampere = 1 coulomb/sec.)

- 39 Prove that if u and v are any real numbers, then

$$|\sin u - \sin v| \leq |u - v|.$$

(Hint: Apply the mean value theorem with $f(x) = \sin x$.)

- 40 Prove that $\sqrt{1+h} < 1 + \frac{1}{2}h$ for $h > 0$. (Hint: Apply the mean value theorem with $f(x) = \sqrt{1+x}$.)

[C] Exer. 41–42: Graph, on the same coordinate axes, $y = f(x)$ for $0 \leq x \leq 1$ and the line through $(0, f(0))$ and $(1, f(1))$. Use the graph to estimate the numbers in $[0, 1]$ that satisfy the conclusion of the mean value theorem.

41 $f(x) = 2.1x^4 - 1.4x^3 + 0.8x^2 - 1$

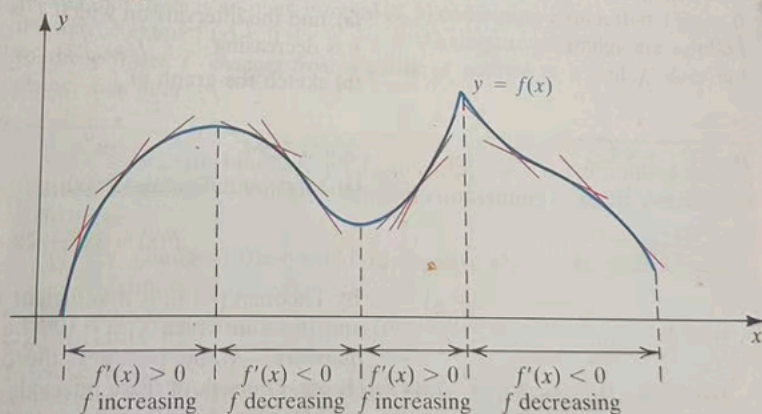
42 $f(x) = \frac{\sin 2x + \cos x}{2 + \cos \pi x}$

4.3 THE FIRST DERIVATIVE TEST

In this section we show how the sign of the derivative f' may be used to determine where a function f is increasing and where it is decreasing. This information will enable us to classify the local extrema of the function.

The graph of $y = f(x)$ in Figure 4.18 indicates that if the slope of the tangent line is positive in an open interval I (that is, if $f'(x) > 0$ for every x in I), then f is increasing on I . Similarly, it appears that if the slope is negative (that is, if $f'(x) < 0$), then f is decreasing.

FIGURE 4.18



That these intuitive observations are actually true is a consequence of the following theorem.

Theorem (4.13)

Let f be continuous on $[a, b]$ and differentiable on (a, b) .

- (i) If $f'(x) > 0$ for every x in (a, b) , then f is increasing on $[a, b]$.
- (ii) If $f'(x) < 0$ for every x in (a, b) , then f is decreasing on $[a, b]$.

PROOF To prove (i), suppose that $f'(x) > 0$ for every x in (a, b) , and consider any numbers x_1, x_2 in $[a, b]$ such that $x_1 < x_2$. We wish to show that $f(x_1) < f(x_2)$. Applying the mean value theorem (4.12) to the interval $[x_1, x_2]$ gives us

$$f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$$